



Guided Work Series N°1

Exercise 01

Which of the following sequences of digits can represent a number in base 16, 8, or 2?

1010110, 230012230, 2546780, 3C52, 8HIE4.

Exercise 02

- What are the **octal** numbers immediately following the numbers 7, 127, 177?
- What are the **hexadecimal** numbers immediately following the numbers: F, 1F9, 2FF?
- What are the numbers in **base 5** immediately following the numbers: 4, 43, 444?

Exercise 03

Convert the following decimal numbers into binary:

- $(256.25)_{10} \rightarrow (?)_2$
- $(88.09)_{10} \rightarrow (?)_2$ (Is there a repeating sequence?)
- $(0.4)_{10} \rightarrow (?)_2$ (Is there a repeating sequence?)

Exercise 04

Convert the following binary numbers into decimal:

- $(10111.1011)_2 \rightarrow (?)_{10}$
- $(110110.0110)_2 \rightarrow (?)_{10}$

Exercise 05

Convert

- $140,5251_7$ to base 10 (use 4 digits precision)
- $23,44_{10}$ to base 5 (use 2 digits precision).
- 0.1_{10} to binary and Hex

Exercise 06

Convert the following decimal numbers into hexadecimal/octal:

- $(89.24)_{10} \rightarrow (?)_{16}$
- $(156)_{10} \rightarrow (?)_{16}$
- $(85.25)_{10} \rightarrow (?)_8$
- $(345)_{10} \rightarrow (?)_8$

Exercise 07

Convert the following hexadecimal/octal numbers into decimal:

- $(234B0.0C)_{16} \rightarrow (?)_{10}$
- $(2A08.EF)_{16} \rightarrow (?)_{10}$
- $(654.05)_8 \rightarrow (?)_{10}$
- $(56.42)_8 \rightarrow (?)_{10}$

Exercise 08

Perform the following conversions:

- $(23B.C)_{16} \rightarrow (?)_8$
- $(76.2F)_{16} \rightarrow (?)_8$
- $(230.5)_8 \rightarrow (?)_{16}$
- $(60.63)_8 \rightarrow (?)_{16}$

Exercise 09

Perform the following operations:

- $(1110.1 + 1011.01)_2$
- $(3CB.A9 + B03.B6)_{16}$
- $(1110.01 - 1011.01)_2$
- $(2476 - 1205.46)_8$
- $(42.23 - 26.15)_6$
- $(1110 \times 110)_2$
- $(B85.23 \times 5)_{16}$
- $(11011 \div 1001)_2$
- $(403 \div 34)_8$